

Automated Medical Podcast Transcription and Topic Segmentation

1. Introduction

In recent years, medical podcasts have become an important source of information for **medical students, healthcare professionals, and researchers**. These podcasts include discussions on diseases, symptoms, treatments, clinical experiences, and public health awareness.

However, medical podcasts are usually **long in duration**, making it difficult for users to quickly find relevant medical topics. Listening to an entire podcast episode to locate a specific disease or treatment discussion is **time-consuming and inefficient**.

To overcome this challenge, there is a need for an **intelligent AI-based system** that can automatically transcribe medical podcast audio, divide it into meaningful topic segments, and provide structured insights such as **summaries, keywords, sentiment analysis, and quality evaluation**.

The **Automated Medical Podcast Transcription and Topic Segmentation** project addresses this problem using **Artificial Intelligence (AI)** and **Natural Language Processing (NLP)** techniques to convert medical audio into **searchable, structured, and user-friendly content**.

2. Problem Statement

Medical podcasts contain valuable clinical and educational information, but their long and unstructured nature makes it difficult for users to:

- Identify specific diseases or treatments quickly
- Navigate topics efficiently
- Analyze the quality and sentiment of discussions

There is no effective mechanism to automatically process and analyze medical podcasts.

Therefore, the problem is to design a system that can:

- Automatically convert medical podcast audio into text
- Detect topic boundaries within the transcript
- Generate summaries, keywords, and sentiment insights
- Evaluate transcription quality

- Provide an interactive user interface
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3. Objectives of the Project

The main objectives of this project are:

1. To convert medical podcast audio into accurate text using speech-to-text models
 2. To segment long transcripts into meaningful medical topics
 3. To extract important medical keywords from each segment
 4. To analyze sentiment in a medical-aware manner
 5. To evaluate transcription quality using WER, CER, and semantic similarity
 6. To provide an interactive frontend for visualization and navigation
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4. Scope of the Project

Included in Scope

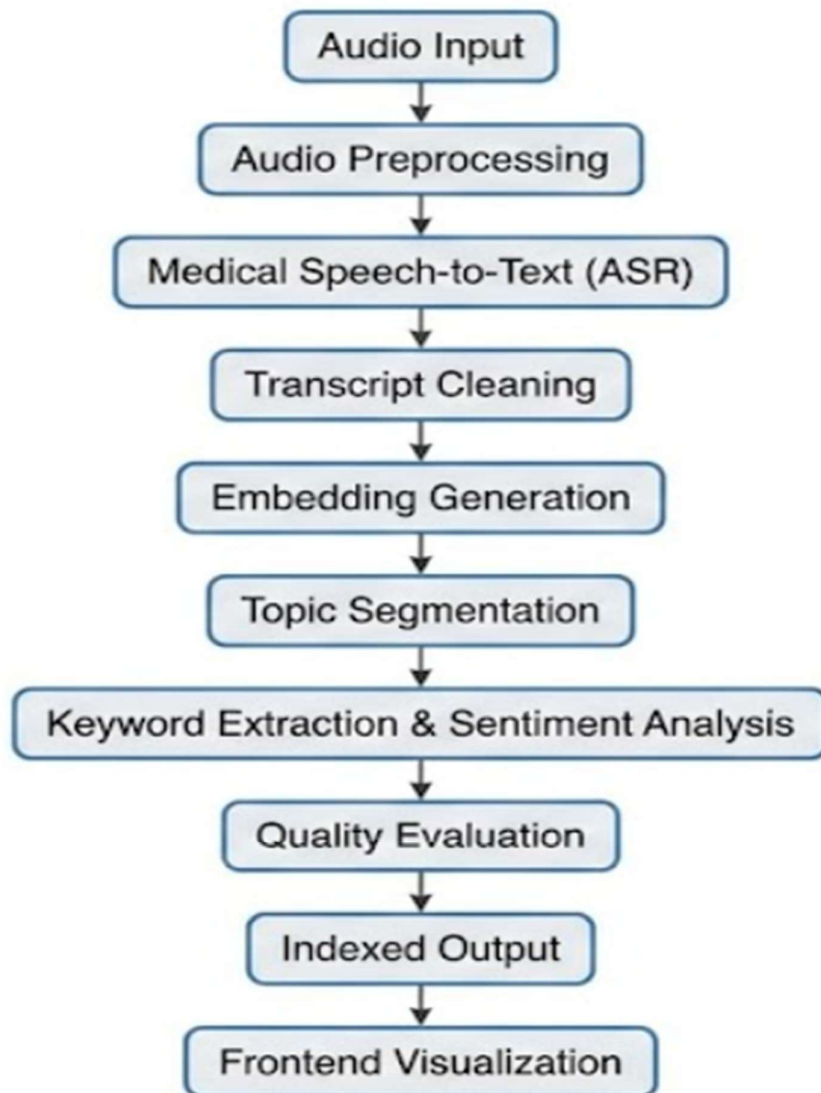
- Processing recorded medical podcast audio files
- Automatic transcription using ASR models
- Topic segmentation using NLP techniques
- Keyword extraction and sentiment analysis
- Quality evaluation using standard metrics
- Visualization through a web-based interface

Excluded from Scope

- Live audio streaming
 - Real-time transcription
 - Clinical diagnosis or treatment recommendation
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5. System Architecture

The system follows a **pipeline-based modular architecture**:



This modular architecture ensures **scalability, easy maintenance, and independent development** of each system component.

6. Module Description

6.1 Audio Preprocessing Module

- Normalizes audio volume
 - Handles background noise and silence
 - Prepares audio for accurate transcription
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6.2 Transcription Module

- Converts audio to text using **Whisper / Faster-Whisper**
 - Generates timestamps for spoken segments
 - Stores transcripts in structured JSON format
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6.3 Topic Segmentation Module

- Uses sentence embeddings
 - Computes similarity between consecutive segments
 - Detects topic change boundaries
 - Produces topic-wise segmented transcripts
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6.4 Keyword Extraction Module

- Extracts important medical terms
 - Uses algorithms such as **KeyBERT / YAKE / RAKE**
 - Helps users understand the topic at a glance
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6.5 Sentiment and Quality Evaluation Module

This module performs:

Sentiment Analysis

- Segment-level sentiment classification
- Medical-aware normalization
- Ensures informative medical discussions are not misclassified as negative
- This is important because serious medical topics are often informative rather than emotionally negative

Quality Evaluation

- Word Error Rate (WER)
 - Character Error Rate (CER)
 - Semantic similarity comparison with clean reference transcripts
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6.6 Frontend Visualization Module

The frontend displays:

- Full transcript
 - Topic-wise summaries
 - Keyword charts
 - Sentiment graphs
 - Quality dashboards
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7. Technology Stack

Backend

- Python 3.9+
- Flask
- Whisper / Faster-Whisper
- Librosa, PyDub, FFmpeg

NLP & Machine Learning

- HuggingFace Transformers
- Sentence Transformers
- NLTK, SpaCy
- KeyBERT / YAKE / RAKE

Frontend

- React.js
- HTML, CSS, JavaScript
- REST APIs

Visualization

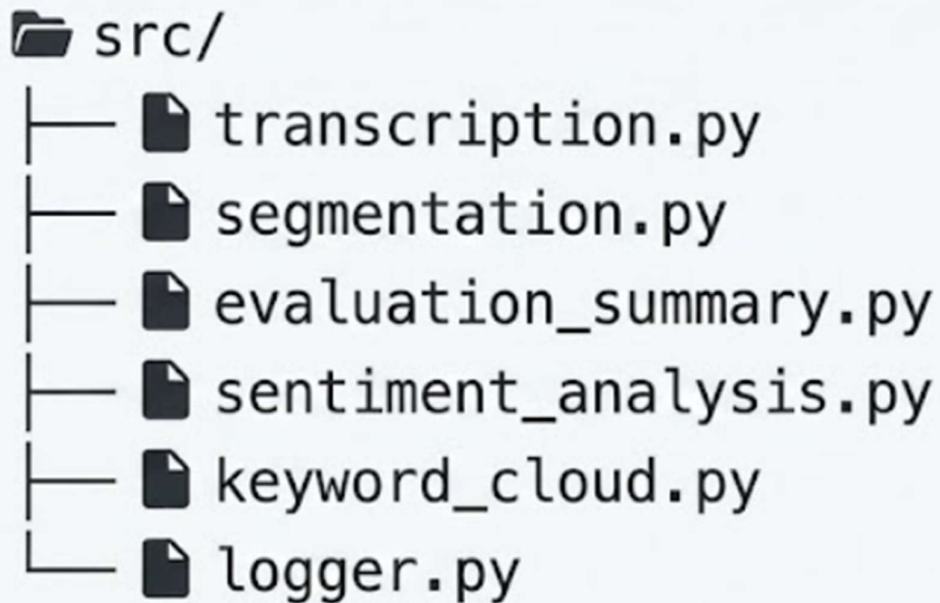
- Chart.js
- Plotly
- Matplotlib

Storage

- JSON, CSV
 - SQLite (optional)
 - FAISS (optional)
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8. Project Structure

8.1 Backend Structure



```
src/  
├── transcription.py  
├── segmentation.py  
├── evaluation_summary.py  
├── sentiment_analysis.py  
├── keyword_cloud.py  
└── logger.py
```

transcription.py

- Performs audio-to-text conversion
- Generates timestamps
- Stores transcript output

segmentation.py

- Topic boundary detection
- Embedding-based similarity analysis

evaluation_summary.py

- Computes WER and CER
- Compares generated transcripts with Clean_transcripts/
- Produces normalized accuracy and similarity scores

sentiment_analysis.py

- Segment-wise sentiment detection
- Medical-domain sentiment normalization

keyword_cloud.py

- Extracts and counts medical keywords

logger.py

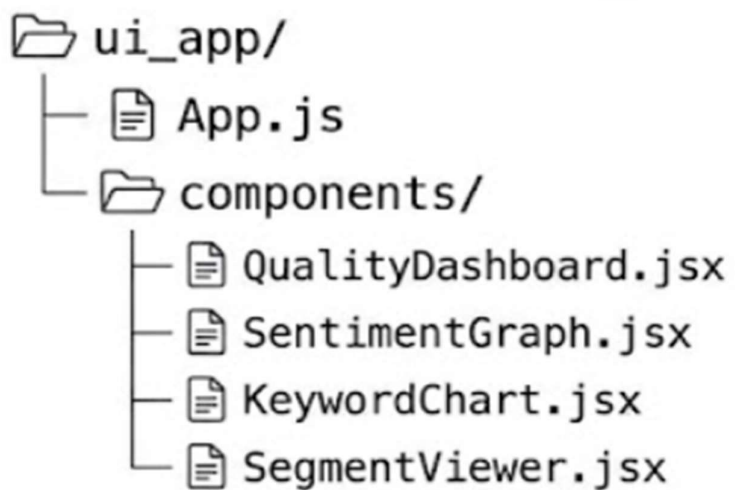
- Centralized logging system
 - Tracks pipeline execution and errors
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8.2 API Layer

app.py

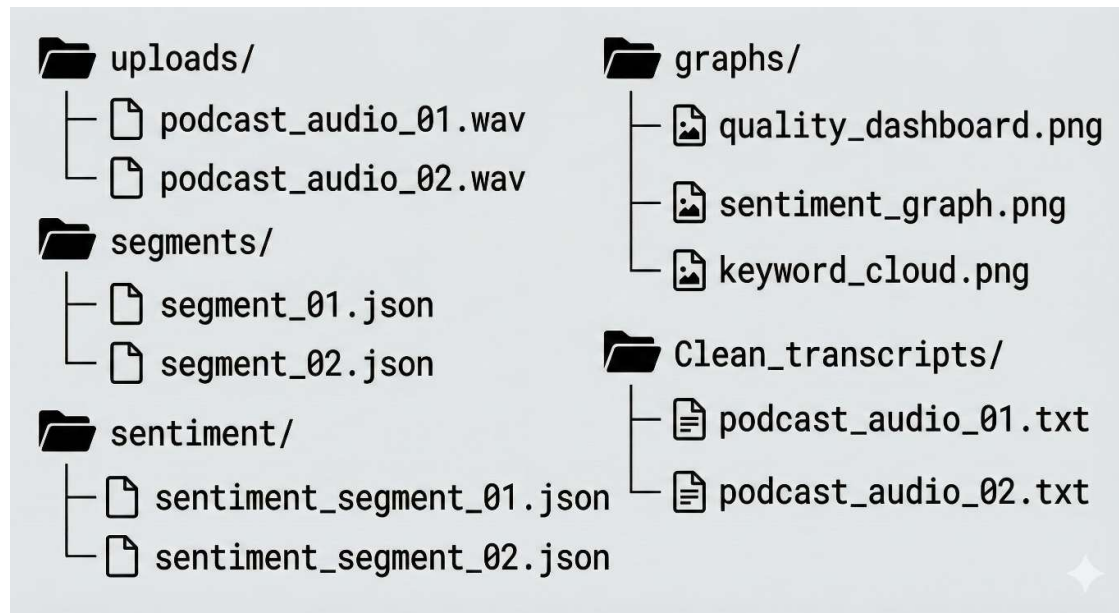
- Flask REST API
 - Handles audio upload and processing
 - Connects backend with frontend
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8.3 Frontend Structure



- Displays dashboards and visual analytics
 - Enables topic navigation using timestamps
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8.4 Data & Output Folders



9. Milestone-wise Implementation

Milestone 1

- Audio preprocessing
- Transcription

Milestone 2

- Topic segmentation
- Keyword extraction

Milestone 3

- Sentiment analysis
- Quality evaluation

Milestone 4

- Frontend integration
- Visualization dashboards

Milestone 5

- Documentation
- Final submission

10. Data and Privacy Considerations

- Audio files are not permanently stored
- API keys are handled securely using environment variables
- No personal user data is stored

11. Future Enhancements

- Medical Named Entity Recognition (NER)
- Speaker diarization
- Semantic search across multiple podcasts
- Multi-language transcription support

12. Intended Users

- Medical students
- Healthcare professionals
- Medical researchers
- Medical educators

13. Conclusion

The **Automated Medical Podcast Transcription and Topic Segmentation** project provides an efficient AI-based solution for analyzing long medical podcasts. By combining **speech recognition, NLP, sentiment analysis, quality evaluation, and visualization**, the system makes medical audio content more accessible, searchable, and informative. This project demonstrates the **practical application of Artificial Intelligence in healthcare education**.

14. License

This project is licensed under the **MIT License**.